

Lab: Explore a Simple Generative Tool

Estimated time needed: 30 minutes

Overview

Generative AI models have revolutionized how you interact with technology, enabling you to create new content, generate realistic images, and translate languages with remarkable accuracy.

In this lab, you will gain hands-on experience with a simple generative AI tool, DataRobot, exploring its capabilities and applications.

Learning Objectives

After completing this lab, you will be able to:

- Sign up in DataRobot
- Add a data set to the use case
- Work on model building

Task 1: Sign-up in DataRobot

Step 1: Click www.datarobot.com

Step 2: Fill in the required information under the "Start your trial today" section and create an account. Once completed the information select "Submit". It will redirect to the signup page.

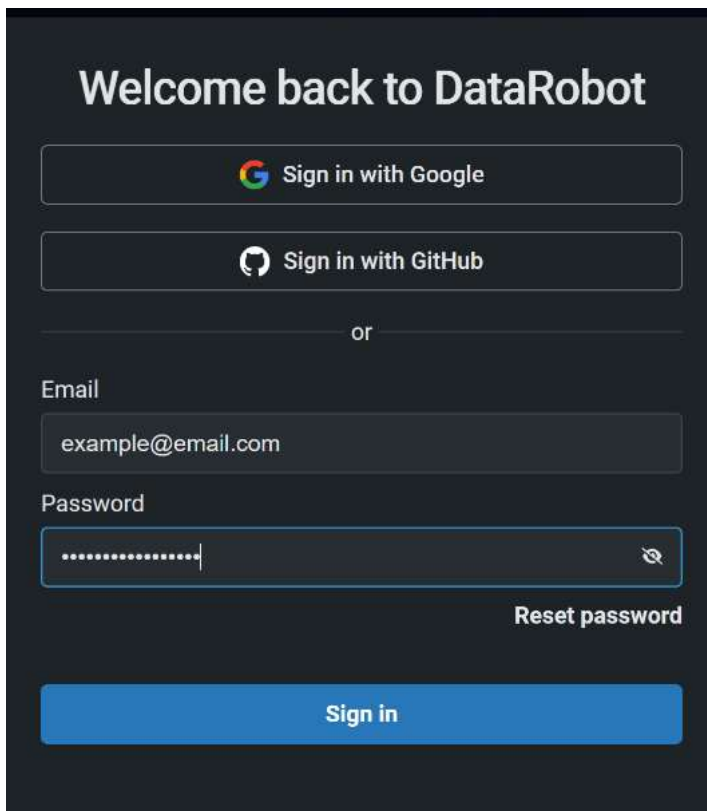
The screenshot shows the DataRobot website. At the top is a navigation bar with the DataRobot logo and links for 'Why DataRobot', 'Apps & Agents', 'Platform', 'Resources', and 'About'. There are also buttons for 'Log In', 'Try DataRobot', and a 'Request Demo' button. Below the navigation bar is a large blue banner with the text 'Try DataRobot for free' and a subtext 'Experience how agents, applications, and an end-to-end suite of production-ready tools can accelerate your production use cases'. Below the banner is a section titled 'Ship a production-grade agent, app, or PoC in 14 days' with a list of bullet points: 'Clone and build the app UI, underlying models, and production-ready components with a single line of code', 'Create a simple agentic workflow by securely connecting to systems of record and adding additional AI tools', and 'Customize all pre-built components based on your specific use case and data in DataRobot hosted notebooks or GUI'. Below this is a video player with the text 'Quickly deliver AI that impacts your business' and 'Easier model delivery'. To the right of the video is a sign-up form titled 'Start your trial today' with fields for 'First Name', 'Last Name', 'Job Title', 'Company', and 'Email Address', each with a placeholder text and an asterisk indicating it is required. There is also a 'Business email' field at the bottom.

Note: To access the DataRobot platform, you must sign up using a work email address. If you do not have a relevant work email, an alternative is to create a GitHub account using your Gmail address. Once registered, you can log in to DataRobot using your GitHub credentials.

For step-by-step guidance on creating a GitHub account, please refer to the following link:

[GitHub Account Setup Guide](#)

Step 3: A new window will open; select the relevant option for signing in.

The image shows the DataRobot login interface. At the top, it says "Welcome back to DataRobot". Below this are two buttons: "Sign in with Google" and "Sign in with GitHub". A horizontal line with the word "or" in the center separates these from the email/password section. The "Email" section has a text input field containing "example@email.com". The "Password" section has a text input field with masked characters (dots) and a toggle icon for visibility. To the right of the password field is a link that says "Reset password". At the bottom is a large blue button labeled "Sign in".

Welcome back to DataRobot

Sign in with Google

Sign in with GitHub

or

Email

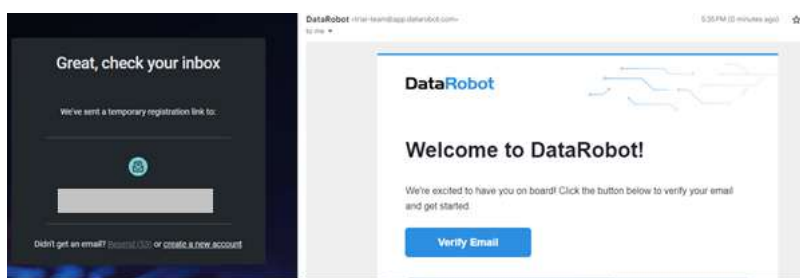
example@email.com

Password

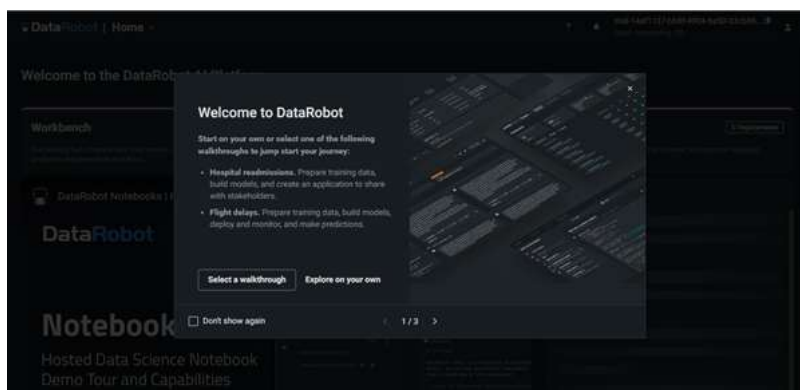
Reset password

Sign in

Step 4: Confirm your email by clicking **Verify Email** in your inbox.



Step 5: Sign up and start your first experience of using the Generative AI tool.
The dashboard will look like the image below. You may like to familiarize yourself with the application by clicking **Select a walkthrough**.



Task 2: Add a data set

Step 8: The dashboard will appear shortly, and your screen will look as shown below. Click **Workbench**.

Shortcuts to AI success
Explore our recommended actions to kickstart your AI journey and unlock the full potential of your data.

Explore application templates
Accelerate development with end-to-end solutions. These code-first, reusable pipelines are available out-of-the-box, but also offer easy customization, for quick, tailored successes.

Build ML and GenAI models
From data to model training to insights, build experiment-based, iterative workflows for predictive and generative AI models.

Recent activity
Workbench Registry Console

Name	Use Case	Type	Last modified	Last i

Step 9: Click **Create Use Case**.

Use Case directory

LAST MODIFIED USE CASES

ALL USE CASES

Name	Datasets	Vector databases	Experiments	Playgrounds	Apps	Notebooks	Created by
F... Walkthrough	1	0	0	0	0	0	P.P.
H... Walkthrough	2	0	0	0	0	0	P.P.

Step 10: Click **Data assets** and select **Browse Data** to include the data set in your use case.

Data

Data associated with this Use Case

There's nothing here yet. Drag-and-drop data here or select an option below to import data for use in experiments and vector databases.

Browse data **Upload file** **Add from URL**

Accepted formats: .csv, .tsv, .dsv, .xls, .xlsx, .sas7bdat, .geojson, .gz, .bz2, .tar, .tgz, .zip, .parquet

[View dataset requirements](#)

Step 10: You can select an in-built sample data set *HOSPITAL_READMISSION_DATA* and select **Add to Use Case**.

Browse data

DataRobot assets

- Data Registry
- Datasets 3

Data connections

Search

+ Add connection

Demo Connection
Snowflake

Data Registry

Source ▾ Owner ▾ Search

☐	Name	Created on ↑	Type	Last m
☒	HOSPITAL_READMISSION_DATA	18m	Snapshot	18m
☐	FLIGHT_DELAYS_SCORING_DATA	18m	Snapshot	18m
☐	FLIGHT_DELAYS_DATA	18m	Snapshot	18m

< Previous page 1 Next page > 3 per page ▾ | 1-3 of 3 items

Step 11: Once you select the data set, you can see a preview of it by clicking on the dataset. You can also view the data set's features, as shown below. Click **Add to Use Case**.

Add data

DATA STORES

Upload Connect

Search

Data Registry
DataRobot

Demo Connection
Snowflake

HOSPITAL_READMISSION_DATA

Data Registry / HOSPITAL_READMISSION_...

Features Data preview

race	gender	age	weight	admission
Categorical	Categorical	Categorical	Categorical	Categorical
Caucasian	Male	[50-60)		Emergency
Caucasian	Female	[70-80)		Urgent
Caucasian	Male	[60-70)		Elective
Caucasian	Male	[60-70)		
Caucasian	Female	[40-50)		Not Availa
Caucasian	Male	[90-100)		Emergency
Caucasian	Male	[70-80)		Urgent
Caucasian	Female	[80-90)		Urgent

Snapshot sample 51 features | 1,511 rows

Step 12: After you add the data set to the use case, the workbench will appear as shown below. You can click the data set to see the feature insights.

Workbench
Workbench directory / **Untitled_4**

?
🔔
trial-c3398e
Days remain

< Use Case directory

USE CASE

Untitled_4
:

Add Data

▼

:: All
Data 1
Vector databases
Experiments
Playgrounds
Applications

Type ▼

🔍 Search

⚙ Settings

Name	Created By	Last Modified ↑	Type	Source	Rows	
HOSPITAL_READMISSION_DATA	Pratiksha V.	now	Snapshot	Snowflake	10000	:

< Previous
1
Next >
1 per page ▼
1-1 of 1 items

Step 13: Explore the **All Features** menu to display specific features.

HOSPITAL_READMISSION_DATA

Jul 29th, 2024 10:39 AM Snapshot ▾ Data actions ▾

Data preview Features Feature lists

Show insights Show features from: All Features ^ + Create feature list

Search	DATAROBOT FEATURE LISTS
All Features	51
Informative Features	40
Raw Features	51

race Categorical	age Categorical	weight Categorical	admission_type_id Categorical	discharge_disposition_id Categorical
Caucasian 54%	[70-80] 26%	==Missing== 96%	Emergency 49%	Discharged 49%
AfricanAmerican 21%	[60-70] 22%	[75-100] 2%	Urgent 19%	Discharged 19%
Other 6%	Other 52%	Other 2%	Other 32%	Other 32%
Caucasian Male	[50-60]		Emergency	Discharged
Caucasian Female	[70-80]		Urgent	Discharged

Snapshot sample | 51 features | 1,511 rows

Task 3: Work on Data Modeling

Step 14: You will have options **Modelling** and **Start wrangling**. You can try data wrangling if you want to. For this lab, you will work on model building under **Data actions** and select **Modelling**. It will take a while to prepare a data set for modelling.

DataRobot | Workbench Workbench directory / Untitled_4 / HOSPITAL_READMISSION_DATA

HOSPITAL_READMISSION_DATA Jul 29th, 2024 10:39 AM Snapshot Data actions ^

Data preview Features Feature lists

Show insights Show features from: All Features + Create feature list

race	gender	age	weight	admission	discharge
Categorical	Categorical	Categorical	Categorical	Categorical	Categorical
Caucasian	Female	[70-80]	==Missing==	Emergency	Discharge
AfricanAmerican	Male	[60-70]	[75-100]	Urgent	Discharge
Other		Other	Other	Other	Other
Caucasian	Male	[50-60]		Emergency	Discharge
Caucasian	Female	[70-80]		Urgent	Discharge

Snapshot sample 51 features | 1,511 rows

Start wrangling
Start modeling
Start feature discovery
Download dataset
Remove dataset

Step 15: Once done, you need to select the **Target feature**. Select **readmitted** as your target feature.

DataRobot | Workbench Workbench directory / Untitled_4

Set up new experiment Dataset Target Additional settings

Target feature
 Select the feature to make predictions on.

readmitted

	Uniq...	Missi...	Mean	Std Dev		
7	9592	-	-	-		
troglitazone	35	Categorical	1	0	-	-
tolbutamide	30	Categorical	2	0	-	-
tolazamide	36	Categorical	2	0	-	-
time_in_hospital	7	Numeric	14	0	4.43	3.021

Experiment
 HOSPITAL
 07-30 11:23

Dataset
 Name

Rows
Features

Target
 No target set

Step 16: The workbench screen will be displayed as shown below. Click **Next**.

DataRobot | Workbench Workbench directory / Untitled_4

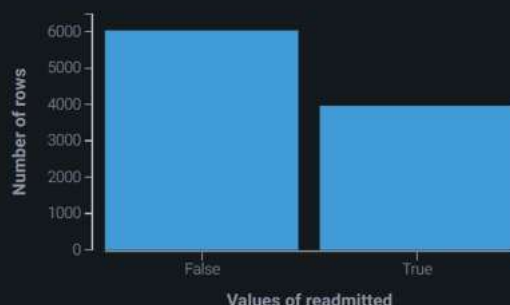
Set up new experiment ✓ Dataset 🎯 Target ⚙️ Additional settings

Target feature
Select the feature to make predictions on.

Target type: Binary classification ⓘ
Positive class: ☐ 0 ☒ 1 ⓘ

Modeling mode
Set the mode used for selecting which blueprints to build when training models.

Optimization metric
Set the metric used when training models to evaluate and optimize accuracy.
 Recommended



Experiment
HOSPITAL_P
07-30 11:23

Dataset
Name

Rows
Features

Target
Feature
Target type
Positive class
Modeling mo
Optimization
Training feat

Partitioning

Step 17: You can modify the model setting in **Additional Settings**; once done, click **Next** and then click **Start modelling**.

DataRobot | Workbench Workbench directory / Untitled_4

Set up new experiment ✓ Dataset ✓ Target 🎯 Additional settings

Data partitioning **Time series modeling** Preview **⚙️ Additional settings**

Partitioning method
Select the method for assigning rows to partitions when training models.

Rows are assigned to ensure similar target distribution across each partition.

Validation type
☒ **Cross-validation**
Trains models on a specified number of folds, maximizing data use but also increasing run time.
☐ **Training-validation-holdout**
Splits data into three partitions: trains models on the training set, assess performance on the validation set, and evaluates the model on unseen data in the holdout set.

Cross-validation folds
Enter a value from 2 - 50.

Holdout percentage
Set the subset of data that is unavailable during training and validation. Enter a value

Experiment
HOSPITAL_P
07-30 11:23

Dataset
Name

Rows
Features

Target
Feature
Target type
Positive class
Modeling mo
Optimization
Training feat

Partitioning

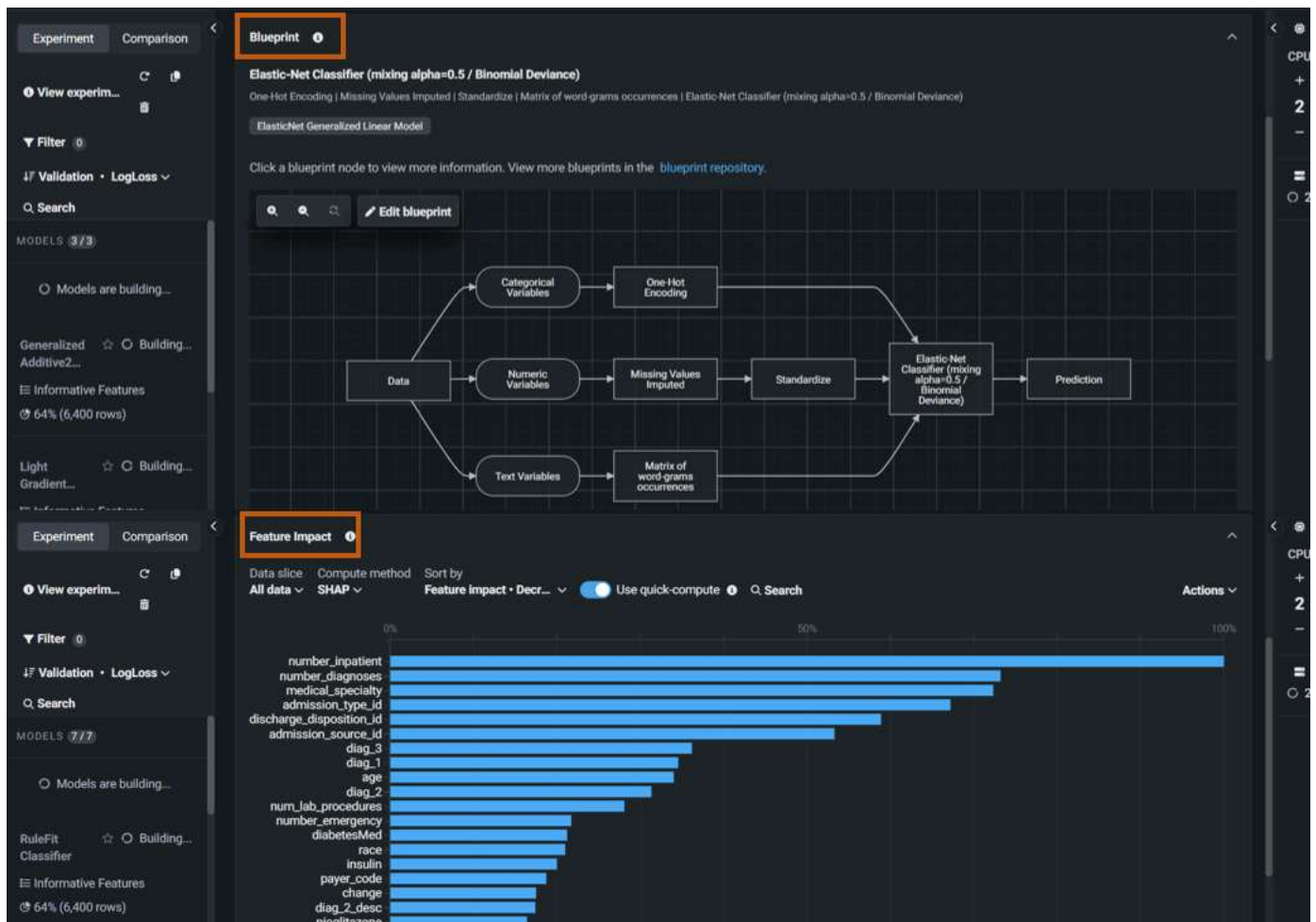
Step 18: Building models will take a while.

The screenshot shows the DataRobot Workbench interface. The top navigation bar includes the DataRobot logo, 'Workbench', and a breadcrumb trail: 'Workbench directory / Untitled_4 / < HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57'. On the left sidebar, there are tabs for 'Experiment' and 'Comparison'. Below these are links for 'View experiment info', a 'Filter' button, and a 'Validation' dropdown set to 'LogLoss'. A search bar is also present. The main panel on the right displays a large circular progress indicator and the text 'Models are building...'. The left sidebar shows a list of models under the heading 'MODELS: 0/0'. Two models are visible: 'Keras Slim Residual Neural Network Classifier using...' and 'Elastic-Net Classifier (L2 / Binomial Deviance)', both marked as 'Building...'.

Step 19: once the modelling is complete, you can pick a model of your choice, and the DataRobot will show the **Model Overview**.

The screenshot shows the 'Model Overview' page in DataRobot Workbench. The top navigation bar is the same as in the previous screenshot. The left sidebar now shows 'MODELS: 3/3'. The main panel is titled 'Model Overview' and displays details for the 'Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)'. The 'Training scores: LogLoss' section shows: Validation (0.6089), Cross-validation (Score), and Holdout (0.6111). The 'Training settings' section shows: Training feature list, Informative Features (64% (6,400 rows)), and Training sample size. Below these are sections for 'Blueprint', 'Feature Impact', 'Feature Effects', 'Individual Prediction Explanations', and 'ROC Curve'. The left sidebar lists three models: 'Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)' with a score of 0.6089, 'Elastic-Net Classifier (L2 / Binomial Deviance)' with a score of 0.6115, and 'Keras Slim Residual Neural' with a score of 0.6327. The first model is highlighted with an orange box.

Step 20: You can explore various model overview components like **Blueprint**, **Feature Impact**, and so on.



Step 21: If you have test or unseen data, you can also make predictions by clicking **Make Predictions** under **Model actions**.

Model Overview

Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)

Training scores: LogLoss

Validation	Score
0.6089	
Cross-validation	0.6111
Holdout	

Training settings

- Training feature list
- Informative Features: 64% (6,400 rows)
- Training sample size

Blueprint

Make Predictions

Make new predictions

Prediction dataset

Prediction datasets cannot exceed 5.00 GB.

Drop file(s) here to upload or [Choose file](#)

Prediction options

- ☒ Include additional feature values in prediction
- ☐ Use model training data
- ☐ Data registry
- ☒ Add all features ☐ Add specified features
-
- ☐ Include Prediction Explanations

[Compute and download predictions](#)

Download recent predictions

Predictions are stored and available to download for 48 h

No recent predictions

Step 22: You can also click **Generate compliance report** and **download compliance report** for your use case.

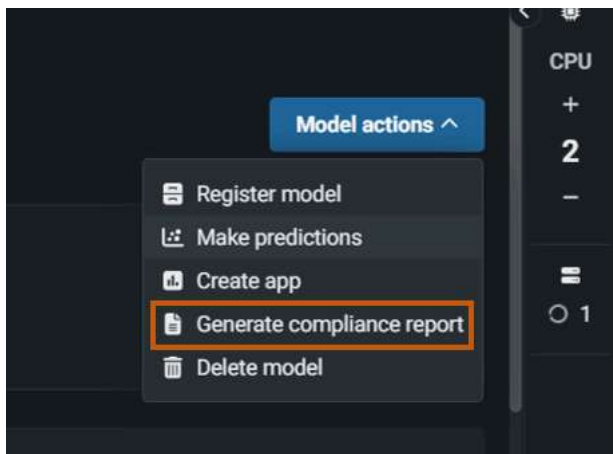


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Conclusion

In this lab, you have signed up in DataRobot, added a data set in a use case, and worked on data modelling.

Author(s)

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